TRACE — Trusted Registry for Autonomous and Connected Entities

A proposed open infrastructure standard for AI agent identity and accountability.

Version 0.3 — June 2026 (Expanded Integrations)

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Executive Summary

AI agents are already operating across organizational and jurisdictional boundaries, performing high-stakes actions without adequate shared infrastructure for identity, authorization, and accountability. TRACE addresses this gap by providing a multi-layer, open standard for trusted agent registries.

This v0.3 expands the Standards Alignment section with explicit mappings to C2PA, OpenTelemetry, Estonia's AI ID initiative, and China's CAC rules, demonstrating how TRACE serves as a practical, complementary backbone.

The Problem
AI agents book meetings, execute financial transactions, control industrial systems, and spawn sub-agents, regardless of organisational and jurisdictional boundaries, with no shared infrastructure to answer a basic question:

> *Who are you, who sent you, and what are you authorised to do?*

Enterprise IAM standards (OAuth 2.0, SPIFFE/SPIRE, OpenID Connect) address agents operating inside a single organisation's trust perimeter. They were not designed for agents crossing that perimeter. No current standard was.

This is not a future risk. It is a present condition.

What TRACE Is

TRACE is a proposed four-layer open standard to close this gap.

Layer	Name	Function
1	Identity	Cryptographic identity bound to a verified legal entity, recorded on a permissioned ledger using W3C DIDs. Recursive Sub-Tokens enable high-velocity delegation without losing chain of custody.
2	Access	Resources query TRACE before granting access. Registered and in good standing: access at the appropriate trust tier. Unregistered: no access, without exception.
3	Governance	The TRACE Foundation — a multi-stakeholder non-profit modelled on ICANN — governs the registry at two speeds: automated Circuit Breakers for real-time containment, human supermajority for permanent revocation.
4	Economic Alignment	High-risk agents post conditional collateral held in escrow. On verified failure, collateral is released to a victim recourse pool. Clean records earn tiered infrastructure incentives.

TRACE complements, not competes with, existing frameworks. It provides the identity substrate the EU AI Act assumes but does not specify; the accountability chain ISO 42001 requires but does not provision; the revocation mechanism NIS2 implies but does not mandate for AI.

Integrations and Interoperability

TRACE is explicitly designed as a **complementary identity and accountability backbone** rather than a replacement for existing tools. The following mappings demonstrate how TRACE strengthens and interoperates with leading standards and initiatives.

C2PA / Content Credentials Integration

Mapping:
TRACE's Identity Layer (W3C DIDs bound to legal entities + recursive sub-tokens) serves as the authoritative **agent signer** for C2PA manifests. When a TRACE-registered agent generates, edits, or acts upon media, code, or other assets:

- The agent's DID (or delegated sub-token) is bound into the C2PA manifest's assertions (e.g., extending `digitalSourceType`, `softwareAgent`, or via proposed custom assertions such as `ai:agentId` and `ai:traceRegistry`).
- Prior to signing, the producing platform or resource performs a real-time TRACE registry lookup to confirm good standing and scoped authorization.
- TRACE Circuit Breakers propagate revocation, invalidating downstream C2PA credentials automatically.

Benefits:
C2PA delivers strong tamper-evident provenance for *what* was created/edited but remains vulnerable to metadata stripping and lacks persistent cross-boundary agent identity. TRACE closes these gaps by enforcing registration at the identity level and providing end-to-end chain-of-custody and revocation. This combination is particularly powerful for high-stakes domains (news, finance, legal, healthcare).

Next Steps: Propose a joint C2PA + TRACE profile for AI agents and contribute to C2PA's AI/ML working groups.

OpenTelemetry Agent Tracing Integration

Mapping:
TRACE's Access and Governance Layers consume and enrich OpenTelemetry (OTel) traces generated by modern agent frameworks:

- Agent spans include the TRACE DID as a `baggage` item or extended semantic convention attribute (building on OpenTelemetry GenAI / AI Agent SIG work).
- Runtime authorization decisions and registry status are recorded in spans.
- Delegation chains (via recursive sub-tokens) are correlated across distributed traces.
- Governance Circuit Breakers use aggregated trace data for real-time anomaly detection and containment.

Benefits:
OpenTelemetry provides rich observability into agent reasoning, tool use, and multi-agent workflows but lacks strong cross-organizational identity, authorization, and revocation. TRACE supplies the missing persistent accountability layer.

Next Steps: Contribute TRACE-specific semantic conventions to the OpenTelemetry specification.

Estonia's AI ID Codes Initiative

Mapping:
TRACE is highly aligned with Estonia's June 2026 launch of official **AI ID codes** — digital identities for agents that are scoped, revocable, and auditable.

- Estonia's national AI IDs map directly onto TRACE DIDs + sub-tokens + permissioned ledger.
- TRACE provides the cross-jurisdictional registry and economic alignment (collateral/escrow) layers that enable Estonian agents to operate reliably beyond national borders.
- Circuit Breakers and governance model complement Estonia's e-Residency and EBSI infrastructure.

Benefits: TRACE operationalizes and scales Estonia's pioneering vision into a global, multi-stakeholder standard while preserving full compatibility.

China's CAC Rules (AI-Generated Content Labeling)

Mapping:
TRACE registration enforces and extends China's CAC requirements for explicit (visible) + implicit (metadata/watermark) labeling of synthetic content:

- Registered TRACE agents automatically embed CAC-compliant labels tied to their verifiable DID.
- Platforms leveraging the TRACE Access Layer can perform registry lookups to verify and enforce labeling.
- Supports CAC's traceability, platform responsibility, and anti-tampering mandates, including in cross-border scenarios.

Benefits: Bridges China's strong domestic content control regime with global interoperability, reducing friction for international agent ecosystems.

Overall Impact: These integrations position TRACE as the unifying "who sent you + revocation" substrate that many existing tools and regulations implicitly assume but do not yet provide.

Standards Alignment

TRACE is designed for compatibility with: W3C DID Core 1.0, OAuth 2.0, NIST SP 800-63-4, NIST AI RMF, ISO/IEC 42001, EU AI Act (Articles 9, 13, 22), NIS2 Directive, EBSI (EU deployments), CSA AICM, **C2PA/Content Credentials**, **OpenTelemetry (GenAI extensions)**, and national initiatives including Estonia AI IDs and China CAC rules.

Status

Milestone	Status
Position Paper v0.3	Draft — June 2026
Position Paper v0.2	Published — March 2026
NIST NCCoE Public Comment	Submitted — March 2026
CSA Blog Submission	Pending
Working Group Formation	Open for interest

Threat Model and Governance (Summary)

[Retain or expand from original paper as needed — include key risks like spoofing, revocation propagation, etc.]

Call to Action

- Join the working group.
- Pilot implementations with agent frameworks.
- Contribute to integrations with C2PA, OTel, etc.
- Provide feedback on this expanded version.
